

Business Case

One of our clients has a sports betting business and manages different brands (websites) for their players in various locations. The csv file "Test Data" found in the same folder as this document contains detailed information of transactions (bets) made by players for a number of matches across different websites, including whether the bets placed were won or lost, the amount of stake placed for each bet, what type of bets did the player make, the odds for each bet placed, etc. In the "Definition" sheet in this excel you will find the data dictionary

Assignment

1. **Create a Sales Summary:** Calculate the company's profit in SGD, and most profitable and least profitable leagues and websites
2. **Share insights from the data:** Based on your analysis, highlight at least 5 insights from the dataset provided. Apply any mathematical methodology that you prefer.
3. **Recommendations:** Propose one idea to help the client increase their profits. Please back up your propositions with quantitative data and statistical analysis.
4. **Predictions:** Predict the Actual_Stake for the English Premier League for each currency for the month of May from the dataset provided. Kindly provide your assumptions and explanations
5. **Prepare a presentation:** Create a presentation explaining your process, your key findings, and your conclusions.

Variable	Type	Definition	Remark
Transaction Id	INT	The identification number of each transaction (bet)	Every transaction ID is unique
TransDate	datetime[μs]	The time when the player placed bet	
UserName	STR	The player's username	
CurrencyStr	STR	Local currency of the transaction	
Actual_Stake	FLOAT	Player's valid stake	This value is in the local currency
Odds	FLOAT	Betting Odds	
Bettype	INT	What was the type of bet placed, in conjunction with Betteam	1: Handicap (favorite vs underdog) 2: Odd/Even (bets on if the total number of goals/points end in a odd or even number) 3: Over/Under (bet on if the quantity of something is more or less than a specific number, usually goals/points) 4: Correct Score (predicting the exact final score of the match specifying the goals for each team) 5: 1X2 (bets if who wins, home team, away team or draw) 6: total goal (the sum of goals at the end of the match, no team differentiation) 7: First Half Handicap (favorite vs underdog) 8: First Half 1X2 (bets if who wins, home team, away team or draw but for the first half) 9: First Half Over/Under (bet on if the quantity of something is more or less than a specific number, usually goals/points but for first half) Example: Betteam: 2 and Bettype: 5 refers to the bet "Away team" has been placed (Player will win the bet only if the Away team wins)

Betteam	STR	What was the bet option placed, in conjunction with Bettype	h:Home Team a:Away Team (h:Over ; a:Under) 1: home team X: draw 2: away team Example: Betteam: h and Bettype: 3 means the bet "Over" has been placed
Status	STR	Status of the player's bet	The status (won, lose, draw) will be considered a valid bet ; D.L void: danger ball bets will be rejected ; VOID: other reason reject bet
Winlost	FLOAT	The amount that the player won or lost (including stake)	This value is in the local currency
ActualRate	FLOAT	Exchange rate of each local currency to SGD	
Winlostdate	DATETIME	Event Date	
LiveIndicator	BOOLEAN	Whether the bet placed was before the match kicked off (Non-Live) or during the match (Live).	0: Non Live ; 1: Live
BetGameTime	STR	At what time during the match the bet was placed	The format includes the Half followed by "H" and then the Minute. So 1H34 means First Half, Minute 34. Only applies to football.
MatchId	INT	Game ID (indicates a specific match)	
SportId	INT	Sport ID (Indicates the sport)	1: Football ; 2: Basketball
LeagueId	INT	League ID (indicates a specific League)	3: English Premier League 4: Italy serie A 5: Germany Bundesliga 6: Spain La Liga 13: Uefa Champions League 86: English Championship 136: Mexico Primera Division 187: NBA 48952: Copa Libertadores 56025: Peru Liga 1 57605: Indonesia Liga 1

WebId	STR	Each WebId represents a Website (brand)	Web_001: BetNova Web_002: WagerZone Web_003: InkaMaxWin Web_004: ColossalWin Web_005: ChasquiPlay Web_006: FastStake Web_007: CuscoOdds Web_008: SureStrike Web_009: PachaWin Web_010: InfinityWin
MaxBet	FLOAT	Maximum amount of stake that a player can place per bet	This value is in the local currency